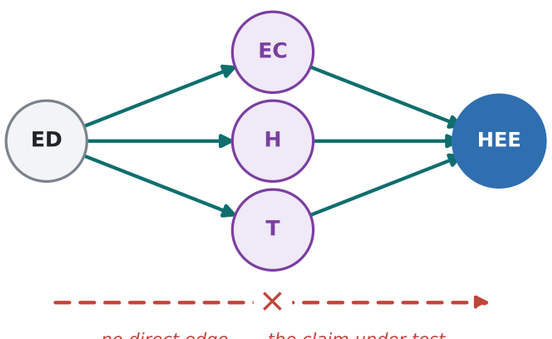
A. One implied independence, tested — the Education exclusion

$\mathcal{A}$ is minimal only if Education reaches HEE entirely through Economic Capital, Housing and Tenure. That is a claim about the data, and the 80-implication battery is 80 tests of exactly this kind.

Education's effect is routed through these three



no direct edge — the claim under test

Upheld

partial r(ED, HEE | D, EC, U) = -0.015
95% CI [-0.026, -0.004], n = 31,802
$p < 0.01$, yet the magnitude is ~0. At this n significance is uninformative and effect size is what the verdict rests on.

Not upheld — and worth saying so

partial r(D, HEE | C, EC, H, U) = -0.270
The full DAG can account for it: conditioning on EC and H opens collider paths into U_{AH}. But the Demographics exclusion is not clean the way the Education exclusion is, and $\mathcal{A}$ rests on both equally.

A test at $n \approx 32{,}000$ rejects almost any exact null, so implications are judged by |partial r|, not p. A failure counts against the graph only if the pair stays d-separated once the unmeasured nodes are restored.

B. What each mark means — and what it does not

mark	appears in	asserts
$X \to Y$	**Expert DAG**	X is assumed to be a direct cause of Y at LSOA scale. *Not a finding. The direction is an assumption defended by literature and review.*
$X \to Y$	**PC output (CPDAG)**	Every DAG consistent with the tested independencies orients it this way. *Not evidence of direction: PC also assumes no unmeasured confounders, which is false here.*
$X - Y$	**PC output (CPDAG)**	X and Y are adjacent; the data do not determine which way. *Not a symmetric or mutual effect.*
$X \leftrightarrow Y$	**FCI output (PAG)**	Neither is an ancestor of the other; an unmeasured common cause drives the association. *NOT mutual causation, NOT feedback, NOT a two-way effect. 28 of 39 FCI edges are of this kind.*
$X \circ\!\!\to Y$	**FCI output (PAG)**	The mark at X is undetermined: the edge may be $X \to Y$ or $X \leftrightarrow Y$. *Not a weaker version of an arrow — an explicit statement of what the data cannot settle.*

One word, three meanings — keep them apart

FCI's ↔ = latent common cause. A "plausibly bidirectional edge" in the orientation-sensitivity analysis = an edge whose direction the experts were unsure of. A "substantively bidirectional" relationship (MH–PH, EC–ED) = genuine feedback that acyclicity forbids.